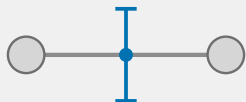


A closed-form error bar turns cycle closure into a quality-control test

1143 real binding edges, 48 systems with a cycle

A per-edge error bar



$$se_e = \sqrt{se_{\text{complex}}^2 + se_{\text{solvent}}^2}$$

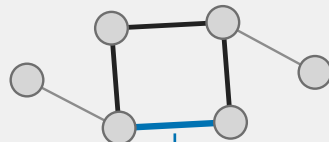
computed, not learned

B validated on replicates

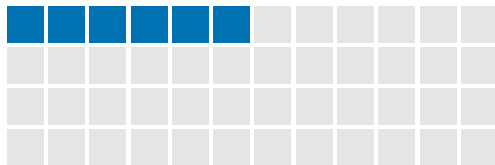


reported/replicate = 1.41

C cycle-closure fit

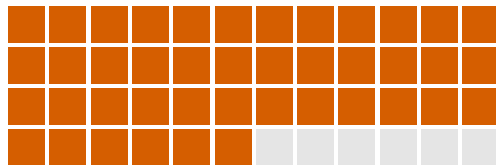


D calibrated



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E overconfident stand-in



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